SAT Math Mastery · Session 6

Percentages, Ratios and Rates

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics percent of a quantity · percent increase and decrease · successive and reverse percent changes · percent error · mixtures and weighted percents · percent chains and parameters · ratios and proportions · rates and direct variation · unit conversion, including squared and cubed units · scale and relative speed · solving with a given total

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type into your calculator or Desmos, one expression per line; for the graphing problem, tap a grey dot to read an intersection. Where a letter such as x stands for another quantity, the step says so.

1. HARD · NO CALCULATOR NEEDED

The table below summarizes the number of students in each of four school clubs.

Club	Number of students
A	500
B	72
C	1,100
D	128
Total	1,800

The number of students in club C is $p\%$ of the number of students in club A. What is the value of p?

1. Club C is $p\%$ of club A, so $p = \dfrac{\text{club C}}{\text{club A}} \times 100$. Only these two clubs matter; ignore B, D and the total.

2. $p = \dfrac{1{,}100}{500} \times 100$, and $\dfrac{1{,}100}{500} = 2.2$.

3. $2.2 \times 100 = 220$. It is over 100 because club C is larger than club A.

Answer: 220

2. HARD · NO CALCULATOR NEEDED

In a factory quality check, 40% of the items inspected in a batch are defective. Of those defective items, 45% are also mislabeled. Which of the following is closest to the percentage of items in the batch that are NOT mislabeled defective items?

(A) 55%

(B) 82% ✓

(C) 96.4%

(D) 18%

1. Take 45% of the 40% that are defective: $0.45 \times 0.4 = 0.18$. So 18% of the whole batch is defective and mislabeled.
2. The question asks for the items that are NOT in that group, so subtract from the whole: $100\% - 18\% = 82\%$.
3. Watch out: 55% is only the share of defective items that are not mislabeled; the question is about the whole batch.

Answer: (B)

3. HARD · NO CALCULATOR NEEDED

The population of a town increased by 10% in the first year and then decreased by 10% in the second year. If the population at the start of the first year was 5,000, what was the population at the end of the second year?

(A) 5,050

(B) 4,900

(C) 4,950 ✓

(D) 5,000

1. A 10% increase multiplies by 1.1: $5{,}000 \times 1.1 = 5{,}500$.
2. The 10% decrease is taken from the new population, 5,500, not from 5,000: $5{,}500 \times 0.9 = 4{,}950$.
3. The decrease removes 550 people but the increase added only 500, so the town ends 50 below where it started, not back at 5,000.

Answer: (C)

4. HARD · CALCULATOR

A renewable-energy report estimated that a new solar farm would generate 480 gigawatt-hours (GWh) of electricity in its first year. The farm actually generated 432 GWh. What is the percent error of the estimate, relative to the actual amount generated?

(A) 11.1% ✓

(B) 12.5%

(C) 9.1%

(D) 10.0%

1. Percent error here is the size of the error divided by the actual amount. The error is $480 - 432 = 48$ gigawatt-hours.
2. Type `48/432*100`: it returns 11.1111, which rounds to 11.1%.
3. Divide by the actual 432, as the question says. Dividing by the estimate 480 instead gives 10%, the tempting wrong choice.

Answer: (A)

5. HARD · NO CALCULATOR NEEDED

The price of an item is increased by 40%, and then the new price is decreased by 25%. The final price is what percent of the original price?

(A) 115% (B) 65%

(C) 100% **(D) 105% ✓**

1. Pick an easy starting price, 100. A 40% increase gives $100 \times 1.4 = 140$.
2. The 25% decrease is taken from the new price: $140 \times 0.75 = 105$.
3. So the final price is 105 out of the original 100, which is 105% of it. Subtracting the percents ($40 - 25 = 15$) would suggest 115%, but the 25% is taken from a larger price.

Answer: (D)

6. HARD · CALCULATOR

A store sells an item at 40% above its cost, then offers a 25% discount. The discounted selling price is \$210. What was the store's cost, in dollars?

1. Let the cost be c. Selling at 40% above cost gives $1.4c$. A 25% discount leaves 75% of that, so the discounted price is $0.75 \times 1.4c = 1.05c$.
2. The discounted price is \$210, so $1.05c = 210$.
3. Type `210/1.05`: it returns 200 dollars. Check: $200 \times 1.4 = 280$ and $280 \times 0.75 = 210$.

Answer: 200

7. HARD · CALCULATOR

A town's population decreased by 20% in 2023 and then increased by 15% in 2024, ending at 9,200. What was the population before the 2023 decrease?

1. Let P be the population before 2023. A 20% decrease multiplies by 0.8 and a 15% increase multiplies by 1.15, so the final population is $P \times 0.8 \times 1.15$.
2. $0.8 \times 1.15 = 0.92$, so $0.92P = 9{,}200$.
3. Type `9200/0.92`: it returns 10,000. Check: $10{,}000 \times 0.8 = 8{,}000$ and $8{,}000 \times 1.15 = 9{,}200$.

Answer: 10000

8. HARD · NO CALCULATOR NEEDED

The number x is 50% greater than y, and y is 20% less than z, where $z > 0$. The number x is what percent of z?

1. Percent questions about ratios work with any positive z, so take $z = 100$. Then y is 20% less than z: $y = 100 - 20 = 80$.

2. x is 50% greater than y: $x = 80 \times 1.5 = 120$.

3. So $x = 120$ when $z = 100$, which means x is 120% of z. (The percents do not simply add: $50 - 20 = 30$ would be wrong.)

Answer: 120

9. HARD · CALCULATOR

A quantity increases by 10%, then increases by 10% again, and then decreases by 20%. By what percent has the quantity decreased from its original value? (Enter as a decimal.)

1. Start with 100. The first 10% increase gives $100 \times 1.1 = 110$.

2. The second 10% increase is taken from 110: $110 \times 1.1 = 121$.

3. Type `100*1.1*1.1*0.8`: it returns 96.8 after the 20% decrease.

4. The quantity went from 100 to 96.8, so it decreased by $100 - 96.8 = 3.2$ percent.

Answer: 3.2

10. HARD · NO CALCULATOR NEEDED

A rectangular banner has an area of 540 square inches. The length of the banner is 150% of its width, and the width is $c\sqrt{10}$ inches, where c is a positive constant. What is the value of c?

1. Let the width be w. The length is 150% of the width, so it is $1.5w$, and the area is $w \times 1.5w = 1.5w^2 = 540$.

2. $w^2 = 540 \div 1.5 = 360$, so $w = \sqrt{360} = \sqrt{36 \times 10} = 6\sqrt{10}$.

3. The width is also $c\sqrt{10}$, so $c = 6$.

Answer: 6

11. HARD · NO CALCULATOR NEEDED

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and m g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of m?

(A) 250 (B) 460

(C) 60 ✓ (D) 100

1. Count the nickel in grams. Alloy X gives $0.2 \times 150 = 30$. Alloy Y gives $0.5m$. The pure nickel gives 40. Total nickel: $30 + 0.5m + 40 = 70 + 0.5m$.
2. The final alloy has mass $150 + m + 40 = 190 + m$ and is 40% nickel, so its nickel is $0.4(190 + m) = 76 + 0.4m$.
3. Set the nickel equal: $70 + 0.5m = 76 + 0.4m$, so $0.1m = 6$ and $m = 60$.
4. Check: nickel is $30 + 30 + 40 = 100$ grams out of 250 grams, and $100 \div 250 = 0.4$, which is 40%.

Answer: (C)

12. HARD · CALCULATOR

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

1. Let g be the yearly growth factor. Three years of growth multiply the area by g^3, and three years give 33.1% growth, so $g^3 = 1.331$.
2. Type `1.331^(1/3)`: it returns 1.1, the cube root.
3. A yearly factor of 1.1 means 10% growth each year. Check: type `1.1^3`: it returns 1.331, which is the 33.1% growth over three years.

Answer: 10

13. HARD · CALCULATOR

Last year a shop sold only mugs and plates. This year the number of mugs sold rose by 30%, the number of plates sold fell by 10%, and the total number of items sold rose by 6%. Of the items sold this year, what percent were mugs, to the nearest tenth of a percent?

(A) 40.0% **(B) 49.1% ✓**

(C) 50.9% (D) 52.0%

1. Let last year's sales be M mugs and P plates. This year: $1.3M$ mugs, $0.9P$ plates, and $1.06(M + P)$ items in total.
2. The totals must agree: $1.3M + 0.9P = 1.06M + 1.06P$, so $0.24M = 0.16P$ and $P = 1.5M$.
3. Take $M = 1$, so $P = 1.5$. This year there are 1.3 mugs, $0.9 \times 1.5 = 1.35$ plates, and 2.65 items in all.
4. Type `1.3/2.65*100`: it returns 49.0566, which is 49.1% to the nearest tenth.

Answer: (B)

14. HARD · CALCULATOR

In 2020, the population of city M was 20% greater than the population of city N. From 2020 to 2024, the population of city M increased by 15% and the population of city N decreased by 8%. The population of city M in 2024 was p% greater than the population of city N in 2024. What is the value of p?

1. Take city N to be 100 in 2020. City M is 20% greater, so M is 120.
2. In 2024, city M grew 15% and city N fell 8%. Type `120*1.15`: it returns 138 for city M, and type `100*0.92`: it returns 92 for city N.
3. M compared with N: type `(138-92)/92*100`: it returns 50, so M is 50% greater than N.

Answer: 50

15. HARD · CALCULATOR

The positive quantities a, b, c and d satisfy the following: b is 0.6% of a, c is 5% of b, and d is 150% of c. If $b + c + d = 27$, what is the value of a?

1. Write everything with a. $b = 0.6\%$ of a is $0.006a$. $c = 5\%$ of b is $0.05 \times 0.006a = 0.0003a$. $d = 150\%$ of c is $1.5 \times 0.0003a = 0.00045a$.
2. Add them: $b + c + d = (0.006 + 0.0003 + 0.00045)a = 0.00675a$, and this equals 27.
3. Type `27/0.00675`: it returns 4,000.

Answer: 4000

16. HARD · NO CALCULATOR NEEDED

A retailer's total sales were the same in March and April. In March, online sales were p percent of the total sales and came to \$18,000. In April, online sales were $(p + 8)$ percent of the total sales and came to \$25,200. What were the retailer's total sales, in dollars, for one of these months?

1. Total sales are the same in both months, so online sales rose from \$18,000 to \$25,200 only because the share rose by 8 percentage points. That rise is 8% of the total T.
2. $25{,}200 - 18{,}000 = 7{,}200$, so $0.08T = 7{,}200$ and $T = 7{,}200 \div 0.08 = 90{,}000$.
3. Check: March online share is $18{,}000 \div 90{,}000 = 20\%$, and April is $25{,}200 \div 90{,}000 = 28\%$, which is 8 more.

Answer: 90000

17. HARD · NO CALCULATOR NEEDED

A chemist has x liters of an acid solution that is 30% acid and mixes in 15 liters of pure acid. She then adds 25 liters of a second acid solution that is 10% acid. The final solution is 35% acid. What is the total volume, in liters, of the final solution?

(A) 125 (B) 70

(C) 85 **(D) 110 ✓**

1. Track the acid. The first solution holds $0.3x$ liters of acid, the pure acid adds 15, and the second solution adds $0.1 \times 25 = 2.5$. Total acid: $0.3x + 17.5$.
2. The final volume is $x + 15 + 25 = x + 40$, and 35% of it is acid: $0.35(x + 40) = 0.35x + 14$.
3. Set the acid equal: $0.3x + 17.5 = 0.35x + 14$, so $0.05x = 3.5$ and $x = 70$.
4. The question asks for the total volume, not x: $70 + 15 + 25 = 110$ liters.

Answer: (D)

18. HARD · CALCULATOR

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

(A) 65% ✓ (B) 15%

(C) 20% (D) 35%

1. Percent changes are easier with a convenient number. Let the current fiction books be 30. After a 20% increase there will be $30 \times 1.2 = 36$.
2. Nonfiction falls 10% to $0.9N$, and that must equal the new fiction count: $0.9N = 36$. Type `36/0.9`: it returns 40, the current number of nonfiction books.
3. Let R be the reference books. Now the total is $30 + 40 + R = 70 + R$. After the changes it is $36 + 36 + R = 72 + R$, which is $1.01(70 + R) = 70.7 + 1.01R$.
4. So $1.3 = 0.01R$ and $R = 130$. The current total is 200 books. Type `130/200*100`: it returns 65, so reference books are 65% of the collection.

Answer: (A)

19. HARD · DESMOS

A store raises the price of a jacket by r percent. Later, it reduces the new price by $(r - 13)$ percent. After both changes, the final price is 10% higher than the original price. What is the value of r?

1. A raise of r percent multiplies the price by $1 + \frac{r}{100}$ and a cut of $(r - 13)$ percent multiplies by $1 - \frac{r - 13}{100}$. Together they give 1.1.
2. Type `y=(1+x/100)(1-(x-13)/100)` and `y=1.1` (letting x stand for r; zoom out to see both crossings).
3. Tap the intersections: $(-12, 1.1)$ and $(25, 1.1)$. A raise needs $r > 0$ (and a cut needs $r - 13 > 0$), so $r = 25$.
4. Check: type `(1+25/100)(1-(25-13)/100)`: it returns 1.1, which matches the target.

Answer: 25

20. HARD · NO CALCULATOR NEEDED

Quantity Q is 60% greater than quantity P. Quantity R is 25% less than Q. Quantity R is 50% greater than quantity S. P is what percent greater than S?

1. Start from S and work upward. Take $S = 100$. R is 50% greater than S: $R = 100 \times 1.5 = 150$.

2. R is 25% less than Q, so $R = 0.75Q$ and $Q = 150 \div 0.75 = 200$.

3. Q is 60% greater than P, so $Q = 1.6P$ and $P = 200 \div 1.6 = 125$.

4. $P = 125$ and $S = 100$, so P is 25% greater than S.

Answer: 25

21. HARD · NO CALCULATOR NEEDED

An 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?

1. Let s be the combined mass of copper and zinc. The tin is half of s, so the whole sample is $s + \dfrac{s}{2} = 1.5s = 84$.

2. $s = 84 \div 1.5 = 56$ grams, so the tin is $56 \div 2 = 28$ grams.

3. Check with the other clue: copper is $2z + 5$, so $3z + 5 = 56$, giving zinc 17 and copper 39. Then $39 + 17 + 28 = 84$.

Answer: 28

22. HARD · NO CALCULATOR NEEDED

For a construction project, a concrete mixture is made from cement, sand, and gravel in the ratio $1 : 2 : 3$ by weight. A batching plant produces gravel at a constant rate of 6 metric tons per hour. At this rate, how many hours will it take the plant to produce 90 metric tons of the mixture?

1. The ratio $1 : 2 : 3$ has $1 + 2 + 3 = 6$ parts, and gravel is 3 of them, which is $\dfrac{3}{6} = 0.5$ of the mixture.

2. If gravel is made at 6 tons per hour and is 0.5 of the mixture, the plant makes mixture at $6 \div 0.5 = 12$ tons per hour.

3. Time $= 90 \div 12 = 7.5$ hours. (Using 6 tons per hour for the whole mixture would give 15, the common slip.)

Answer: 7.5

23. HARD · CALCULATOR

Car A travels 240 kilometers in 3 hours at a constant speed, and Car B travels 350 kilometers in 4.5 hours at a constant speed. Which of the following is closest to how much faster, in kilometers per hour, Car A's speed is compared with Car B's speed?

(A) 0.9

(B) 2.2 ✓

(C) 4.4

(D) 10.0

1. Speed is distance divided by time. Car A: type `240/3`: it returns 80 kilometers per hour.
2. Car B: type `350/4.5`: it returns 77.7778 kilometers per hour.
3. How much faster A is: type `240/3-350/4.5`: it returns 2.2222, which is closest to 2.2.

Answer: (B)

24. HARD · NO CALCULATOR NEEDED

A metal block has a mass of 540 grams and a volume of 60 cubic centimeters. What is the density of the block, in kilograms per cubic meter? (1 gram per cubic centimeter = 1,000 kilograms per cubic meter.)

(A) 90

(B) 900

(C) 9,000 ✓

(D) 9

1. Density is mass divided by volume: $540 \div 60 = 9$ grams per cubic centimeter.
2. The problem says 1 gram per cubic centimeter equals 1,000 kilograms per cubic meter, so multiply by 1,000: $9 \times 1{,}000 = 9{,}000$.
3. Sanity check: water is 1,000 kilograms per cubic meter and this metal is 9 times as dense, so 9,000 is reasonable; 9 is just the unconverted number.

Answer: (C)

25. HARD · CALCULATOR

A water treatment plant produces fresh water at a constant rate of 1,250,000 liters per hour. At this rate, how many cubic meters of water does the plant produce in one day? (1 cubic meter = 1,000 liters.)

1. One day has 24 hours, so multiply the hourly rate by 24. Type `1250000*24`: it returns 30,000,000 liters.
2. There are 1,000 liters in a cubic meter, so divide by 1,000. Type `30000000/1000`: it returns 30,000 cubic meters.
3. Check: 1,250,000 liters per hour is 1,250 cubic meters per hour, and $1{,}250 \times 24 = 30{,}000$.

Answer: 30000

26. HARD · NO CALCULATOR NEEDED

The value of y varies directly with x. When $x = 9$, $y = 45$. What is the value of y when $x = 14$?

(A) 70 ✓ (B) 77

(C) 56 (D) 63

1. Direct variation means $y = kx$ for a constant k.
2. Use $x = 9,\ y = 45$ to find k: $k = 45 \div 9 = 5$.
3. Now $y = 5 \times 14 = 70$ when $x = 14$.

Answer: (A)

27. HARD · NO CALCULATOR NEEDED

A bus rapid transit line carries passengers at a constant average rate. The line carries 15,000 passengers over a 6-hour morning service window. At this same constant rate, which of the following is closest to the number of passengers carried during a 2.5-hour afternoon service window?

(A) 7,500 (B) 3,750

(C) 5,000 **(D)** 6,250 ✓

1. Find the rate: $15{,}000 \div 6 = 2{,}500$ passengers per hour.
2. Multiply by the 2.5-hour window: $2{,}500 \times 2.5 = 6{,}250$ passengers.
3. Sanity check: 2.5 hours is a bit less than half of 6 hours, so the answer should be a bit less than half of 15,000, which is 7,500. It is.

Answer: (D)

28. HARD · NO CALCULATOR NEEDED

Two cars are racing at constant speeds around a circular track. Car A requires 12 seconds to travel once around the track, and Car B requires 20 seconds to travel once around the track. If Car A passes Car B at a certain moment, how many seconds will elapse before Car A next passes Car B?

1. Count laps. In t seconds Car A completes $\frac{t}{12}$ laps and Car B completes $\frac{t}{20}$ laps. Car A passes Car B again when it has gained exactly one full lap.
2. Gain per second: $\frac{1}{12} - \frac{1}{20} = \frac{5}{60} - \frac{3}{60} = \frac{2}{60} = \frac{1}{30}$ lap per second.
3. One lap of gain takes $1 \div \frac{1}{30} = 30$ seconds. (The number 60 is when both cars are back at the start together, not when A next passes B.)

Answer: 30

29. HARD · NO CALCULATOR NEEDED

A pump fills a rectangular tank at a rate of 0.8 cubic meter per minute. The tank is 12 meters long, 5 meters wide and 2 meters deep. How many hours does it take to fill the tank completely? (Enter as a decimal.)

1. Volume of the tank: $12 \times 5 \times 2 = 120$ cubic meters.

2. Time = volume ÷ rate = $120 \div 0.8 = 150$ minutes.

3. The answer is in hours: $150 \div 60 = 2.5$.

Answer: 2.5

30. HARD · CALCULATOR

A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile = 5280 feet)

(A) $\frac{1}{4}$ (B) $\frac{1}{240}$

(C) 15 ✓ (D) 79,200

1. Convert one unit at a time. Feet to miles: divide by 5,280. Seconds squared to minutes squared: one minute is 60 seconds, so one second squared is $\frac{1}{60^2}$ of a minute squared, and the number gets multiplied by $60^2 = 3{,}600$.

2. Type `22*3600/5280` : it returns 15, so 22 feet per second squared is 15 miles per minute squared.

3. Slips to avoid: multiplying by 3,600 but forgetting to divide by 5,280 gives 79,200; dividing only by 5,280 gives $\frac{22}{5{,}280} = \frac{1}{240}$.

Answer: (C)

31. HARD · CALCULATOR

A car's fuel economy is 40 miles per gallon. Use 1 mile = 1.6 kilometers and 1 gallon = 3.8 liters. How many liters of fuel does the car need to drive 304 kilometers?

1. Work in kilometers. 40 miles per gallon is $40 \times 1.6 = 64$ kilometers per gallon.

2. Gallons needed for 304 kilometers: $304 \div 64 = 4.75$ gallons.

3. Convert to liters at 3.8 liters per gallon in one line: type `304/(40*1.6)*3.8` : it returns 18.05 liters.

Answer: 18.05

32. HARD · CALCULATOR

A region has a population density of 1,250 people per square kilometer. Using 1 mile = 1.6 kilometers, how many people are there per square mile?

1. A square mile is a square that is 1 mile on each side, so it measures 1.6 km by 1.6 km. Type `1.6^2`: it returns 2.56 square kilometers.
2. Each square kilometer holds 1,250 people, so one square mile holds type `1250*2.56`: it returns 3,200 people.
3. Squared unit, so use 1.6^2, not 1.6. Multiplying by only 1.6 would give 2,000.

Answer: 3200

33. HARD · NO CALCULATOR NEEDED

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A?

1. Force = pressure × area. At first each of the 4 legs carries $12A$ newtons, so the table and box weigh $4 \times 12A = 48A$ newtons.
2. After the extra 120 newtons, the total is $48A + 120$, shared by 3 legs that each carry $24A$: $3 \times 24A = 72A$.
3. Set them equal: $48A + 120 = 72A$, so $120 = 24A$ and $A = 5$.

Answer: 5

34. HARD · NO CALCULATOR NEEDED

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

1. Blue to not-blue is 3 : 11, so blue is 3 of every $3 + 11 = 14$ equal parts of the 140 drives. One part is $140 \div 14 = 10$, so blue is $3 \times 10 = 30$ and not-blue is 110.
2. The 110 non-blue drives are red and green in the ratio 7 : 4, which is $7 + 4 = 11$ parts. One part is $110 \div 11 = 10$, so red is $7 \times 10 = 70$ and green is $4 \times 10 = 40$.
3. Red minus blue is $70 - 30 = 40$.

Answer: 40

35. HARD · CALCULATOR

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare = 10,000 square meters.)

(A) 384

(B) 3.84 ✓

(C) 6

(D) 38.4

1. The copy is 125% of the plan, so every length on the copy is 1.25 times its length on the plan. Thus 1.25 cm on the copy is 40 m of ground, and 1 cm on the copy is: type `40/1.25`: it returns 32 meters.
2. The parcel measures 7.5 cm by 5 cm on the copy. Its length: type `7.5*32`: it returns 240 meters. Its width: type `5*32`: it returns 160 meters.
3. Area: type `240*160`: it returns 38,400 square meters. Hectares: type `38400/10000`: it returns 3.84 hectares.
4. Slip to avoid: using 1 cm = 40 m without allowing for the enlargement gives 6 hectares.

Answer: (B)

PART B · SELF-ASSESSMENT — 22 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

The number m is 24% greater than 75. If m is the product of 75 and r, what is the value of r?

1. "24% greater than 75" means 75 plus 24% of 75: $m = 75 + 0.24(75)$.

2. Factor out the 75: $m = 75(1 + 0.24) = 75(1.24)$.

3. The problem says $m = 75r$, so $r = 1.24$. Watch out: 0.24 is only the increase, not the full multiplier.

Answer: 1.24

37. MEDIUM · CALCULATOR

A household's monthly electricity bill was \$340.00. After a rate adjustment, the bill is now \$378.20. To the nearest tenth of a percent, by what percent did the amount of the bill increase?

(A) 11.9%

(B) 12.4%

(C) 10.1%

(D) 11.2% ✓

1. Percent increase $= \dfrac{\text{new} - \text{old}}{\text{old}} \times 100$. The increase is $378.2 - 340 = 38.2$ dollars.

2. Type `38.2/340*100`: it returns 11.2353, which is 11.2% to the nearest tenth.

3. Divide by the old bill, 340, not the new one. Dividing by 378.2 gives 10.1% instead.

Answer: (D)

38. MEDIUM · CALCULATOR

Marcus purchased a car for \$90,000. He estimates that each year the value of the car will depreciate by 15% of the previous year's estimated value. What is the estimated value of the car exactly 2 years after Marcus purchased it?

(A) \$65,025 ✓

(B) \$76,500

(C) \$61,200

(D) \$63,750

1. Losing 15% each year means the value is multiplied by 0.85 each year, so after 2 years it is $90{,}000 \times 0.85^2$.

2. Type `90000*0.85^2`: it returns 65,025 dollars.

3. One year gives only 76,500 dollars, which is a different answer choice. The second year's loss is 15% of 76,500, not of 90,000.

Answer: (A)

39. Medium · Calculator

A school's enrollment decreased from 720 students to 640 students over one year. What is the percent decrease in enrollment?

(A) 10%

(B) 11.1% ✓

(C) 12.5%

(D) 14.3%

1. Percent decrease $= \dfrac{\text{old} - \text{new}}{\text{old}} \times 100$. The enrollment dropped by $720 - 640 = 80$ students.
2. Type `80/720*100`: it returns 11.1111, which is about 11.1%.
3. Divide by the starting value, 720. Dividing by 640 gives 12.5%, the tempting wrong choice.

Answer: (B)

40. Medium · Calculator

A laptop has a listed price of \$3,200. A store offers a 15% discount, and then a 15% sales tax is added to the discounted price. What is the final price of the laptop, in dollars?

(A) 2,976

(B) 3,008

(C) 3,128 ✓

(D) 3,200

1. Discount first: paying 85% of the listed price. Type `3200*0.85`: it returns 2,720 dollars.
2. Now add 15% tax to the discounted price: type `2720*1.15`: it returns 3,128 dollars.
3. The tax is taken from 2,720, a smaller base than 3,200, so the final price is below 3,200. In one line: $0.85 \times 1.15 = 0.9775$, which is 97.75% of the listed price.

Answer: (C)

41. Medium · No calculator needed

After a 20% discount, the price of a jacket is \$256. What was the original price, in dollars, of the jacket before the discount?

1. After a 20% discount you pay 80% of the original price p, so $0.8p = 256$.
2. $p = 256 \div 0.8 = 320$ dollars.
3. Check: 20% of 320 is 64 and $320 - 64 = 256$. (Adding 20% to 256 would give 307.2, which is wrong because the discount was taken from the original price.)

Answer: 320

42. Medium · Calculator

A survey of 850 eleventh-grade students found that 34% plan to take the SAT this year. Which of the following is closest to the number of students who plan to take the SAT?

(A) 289 ✓ (B) 306

(C) 255 (D) 272

1. Find 34% of 850. Type `0.34*850`: it returns 289 students.
2. Check by pieces: 10% of 850 is 85, so 30% is 255, and 4% is 34. Then $255 + 34 = 289$.
3. Sanity check: 34% is a little more than one third, and one third of 850 is about 283.3, so 289 students fits.

Answer: (A)

43. Medium · No calculator needed

If 30 percent of 40 percent of x is equal to 50 percent of y minus 20 percent of y, what is the value of x in terms of y?

(A) $x = 3y$ (B) $x = 5y$

(C) $x = 2y$ **(D)** $x = 2.5y$ ✓

1. "Of" means multiply, so 30% of 40% of x is $0.3 \times 0.4\,x = 0.12x$.
2. Right side: 50% of y minus 20% of y is $0.5y - 0.2y = 0.3y$.
3. $0.12x = 0.3y$, so $x = \dfrac{0.3}{0.12}\,y = 2.5y$.

Answer: (D)

44. Medium · No calculator needed

The mass of object A is k grams. The mass of object B is 130% of the mass of object A. If the mass of object B is 66 grams greater than the mass of object A, what is the value of k?

1. Object B has mass $1.3k$ and is 66 grams more than A, so $1.3k = k + 66$.
2. Subtract k: $0.3k = 66$, so $k = 66 \div 0.3 = 220$.
3. Check: B is $1.3 \times 220 = 286$, and $286 - 220 = 66$.

Answer: 220

45. Medium · Calculator

A lab combines 40 grams of a solution that is 0.3% sodium chloride by mass with 60 grams of a solution that is 0.05% sodium chloride by mass. What is the percent of sodium chloride by mass in the combined 100 grams of mixture?

1. Find the grams of sodium chloride in each solution. First: type `0.003*40`: it returns 0.12 grams (that is 0.3% of 40).
2. Second: type `0.0005*60`: it returns 0.03 grams (that is 0.05% of 60).
3. Together: $0.12 + 0.03 = 0.15$ grams of salt in 100 grams of mixture, so the mixture is 0.15% sodium chloride. (Averaging 0.3 and 0.05 to get 0.175 ignores that the amounts differ.)

Answer: 0.15

46. Medium · Calculator

A car consumes fuel at a rate of 6.5 liters per 100 kilometers. At this rate, how many liters of fuel are needed to travel 340 kilometers?

(A) 20.4 (B) 21.0

(C) 22.1 ✓ (D) 23.9

1. The rate is 6.5 liters for every 100 kilometers, which is 0.065 liter per kilometer.

2. For 340 kilometers: type `6.5/100*340` : it returns 22.1 liters.

3. Check by scaling: 340 km is 3.4 times 100 km. Type `6.5*3.4` : it returns 22.1, the same amount.

Answer: (C)

47. Medium · No calculator needed

A recipe for granola bars requires oats, honey, and raisins in the ratio 5 : 2 : 3 by weight. If a bakery uses 18 kilograms of raisins to prepare a large batch, how many kilograms of oats are used?

(A) 24 **(B) 30 ✓**

(C) 36 (D) 12

1. Raisins are 3 parts of the ratio 5 : 2 : 3 (oats : honey : raisins) and weigh 18 kilograms.

2. One part is $18 \div 3 = 6$ kilograms.

3. Oats are 5 parts: $5 \times 6 = 30$ kilograms.

Answer: (B)

48. Medium · No calculator needed

A car travels at a constant speed of 90 kilometers per hour. What is this speed in meters per second?

(A) 32.4 (B) 324

(C) 15 **(D) 25 ✓**

1. Use 1 km = 1,000 m and 1 hour = 3,600 s. Then 90 km per hour is 90,000 m per 3,600 s.

2. $90{,}000 \div 3{,}600 = 25$ meters per second.

3. Shortcut to remember: kilometers per hour divided by 3.6 gives meters per second, and $90 \div 3.6 = 25$.

Answer: (D)

49. Medium · No calculator needed

A recipe that serves 8 people uses 3 cups of orange juice and 5 cups of water. If the same ratio of ingredients is used to make a batch that serves 24 people, how many cups of orange juice are needed?

(A) 9 ✓ (B) 15

(C) 6 (D) 7.5

1. The new batch serves $24 \div 8 = 3$ times as many people, so every ingredient is multiplied by 3.

2. Orange juice: $3 \times 3 = 9$ cups.

3. The water would be $5 \times 3 = 15$ cups, which is a different answer choice. The question asks for orange juice.

Answer: (A)

50. Medium · No calculator needed

A factory produces cardboard boxes at a constant rate of 480 boxes every 8 hours. At this rate, how many boxes does the factory produce in 5.5 hours?

1. Unit rate: $480 \div 8 = 60$ boxes per hour.

2. For 5.5 hours: $60 \times 5.5 = 330$ boxes.

3. Check in pieces: 5 hours is 300 boxes and half an hour is 30 boxes, and $300 + 30 = 330$.

Answer: 330

51. Medium · No calculator needed

An architect built a scale model of a planned tower, where every 1 centimeter on the model represents 2.5 meters of the actual tower. If the model is 18 centimeters tall, what is the height, in meters, of the actual tower?

(A) 36 **(B)** 45 ✓

(C) 50 (D) 20

1. Every 1 cm on the model is 2.5 m on the real tower, so multiply the model height by 2.5.

2. $18 \times 2.5 = 45$ meters.

3. Check in pieces: 10 cm is 25 m and 8 cm is 20 m, and $25 + 20 = 45$.

Answer: (B)

52. Medium · No calculator needed

A saline solution is mixed so that the ratio of salt to water is 2 to 15. If a container holds 51 liters of the solution in total, how many liters of salt are in the container?

(A) 12 (B) 3

(C) 6 ✓ (D) 9

1. Salt to water is $2 : 15$, so the 51 liters of solution are split into $2 + 15 = 17$ equal parts.

2. One part is $51 \div 17 = 3$ liters.

3. Salt is 2 parts: $2 \times 3 = 6$ liters. (The water is $15 \times 3 = 45$ liters, and 3 liters is just one part.)

Answer: (C)

53. Medium · No calculator needed

A gift box contains almonds and cashews, in a ratio of 7 to 5. When 8 almonds and 8 cashews are removed from the box, the ratio of almonds to cashews becomes 3 to 2. How many cashews were originally in the box?

(A) 40 ✓ (B) 48

(C) 24 (D) 32

1. Write the amounts as $7k$ almonds and $5k$ cashews, since the ratio is $7 : 5$.

2. After removing 8 of each, $\dfrac{7k-8}{5k-8} = \dfrac{3}{2}$. Cross-multiply: $2(7k-8) = 3(5k-8)$, so $14k - 16 = 15k - 24$.

3. Solve: $24 - 16 = 15k - 14k$, so $k = 8$.

4. Cashews at the start: $5k = 40$. Check: $56 - 8 = 48$ almonds and $40 - 8 = 32$ cashews, and $48 \div 32 = \dfrac{3}{2}$.

Answer: (A)

54. Medium · Calculator

A rectangular banner is 2.4 meters by 1.5 meters. Printing costs \$0.05 per square centimeter. What is the cost, in dollars, to print the banner? (1 meter = 100 centimeters.)

1. The price is per square centimeter, so change the sides to centimeters first: 2.4 m = 240 cm and 1.5 m = 150 cm.

2. Area: type `240*150`: it returns 36,000 square centimeters.

3. Cost: type `36000*0.05`: it returns 1,800 dollars. Slip to avoid: one square meter is $100 \times 100 = 10{,}000$ square centimeters, not 100.

Answer: 1800

55. Medium · Calculator

A train travels at a constant speed of 45 miles per hour. What is the train's speed, in feet per second? (1 mile = 5280 feet)

(A) 3,960 (B) 237,600

(C) 12.5 **(D) 66 ✓**

1. Change miles to feet and hours to seconds: 45 miles per hour is $45 \times 5{,}280$ feet per 3,600 seconds.
2. Type `45*5280/3600`: it returns 66 feet per second.
3. Check with a landmark: 60 miles per hour is exactly 88 feet per second, and $88 \times \frac{45}{60} = 66$. Forgetting to divide by 3,600 would give 237,600.

Answer: (D)

56. Medium · No calculator needed

A rectangular storage box is 0.4 meter long, 0.5 meter wide and 0.3 meter tall. What is the volume of the box, in cubic centimeters? (1 meter = 100 centimeters.)

1. Volume in cubic meters: $0.4 \times 0.5 \times 0.3 = 0.06$.
2. One cubic meter is $100 \times 100 \times 100 = 1{,}000{,}000$ cubic centimeters, because all three lengths are converted.
3. $0.06 \times 1{,}000{,}000 = 60{,}000$ cubic centimeters. Check by converting the sides first: $40 \times 50 \times 30 = 60{,}000$.

Answer: 60000

57. Hard · No calculator needed

A rectangular banner has width w and length $1.8w$, where $w > 0$. The area of the banner equals the combined area of 4 identical square tiles. The side length of each tile is $k\,w\sqrt{5}$, where k is a positive constant. What is the value of k?

1. The banner's area is $w \times 1.8w = 1.8w^2$.
2. Each tile is a square with side $kw\sqrt{5}$, so its area is $(kw\sqrt{5})^2 = 5k^2w^2$, and 4 tiles have area $20k^2w^2$.
3. Set the areas equal: $20k^2w^2 = 1.8w^2$, so $k^2 = 1.8 \div 20 = 0.09$ and $k = 0.3$, taking the positive root.

Answer: 0.3